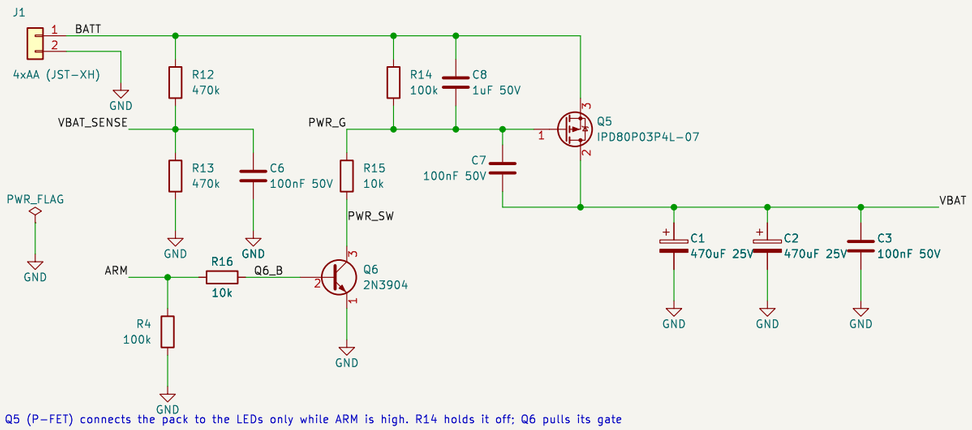
LITE control board — LEDs from a 4 x AA NiMH pack; keys straight to ESP32 GPIOs

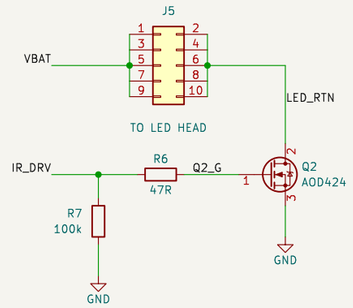
The LEDs light only while ARM (GPIO2) powers them through Q5 AND the carrier on IR_DRV (GPIO17) switches Q2. LED head: 33 strings of 2 x SFH 4545 + 24R.

A. Battery, pack sense and LED power gate



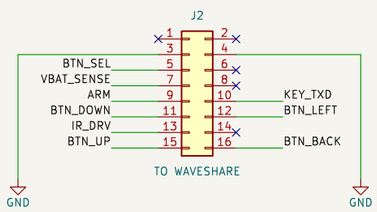
Q5 (P-FET) connects the pack to the LEDs only while ARM is high. R14 holds it off; Q6 pulls its gate down through R15 (Vgs = −0.91 x pack). C7 (gate-drain) is a soft start, C8 (gate-source) keeps Q5 off when cells go in. R4 holds ARM low while the ESP32 boots. Pack sense: BATT / 2 (470k + 470k, 5 uA) on GPIO1.

B. LED switch (Q2) and LED head connector (J5)

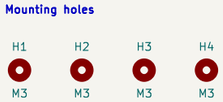


J5 -> LED head J1 over the 10-way ribbon: odd pins VBAT (LED_PWR), even pins LED_RTN.
Q2 is logic-level (5.7 mOhm max at Vgs 2.5 V): GPIO17 drives it directly. R6 47R slows its edge so the ribbon can't ring the LEDs into reverse bias (sim/). R7 holds Q2 off while the ESP32 boots.

C. J2 — Waveshare ESP32-S3-RLCD-4.2 2x8 header (16-way ribbon, pin n = pin n)

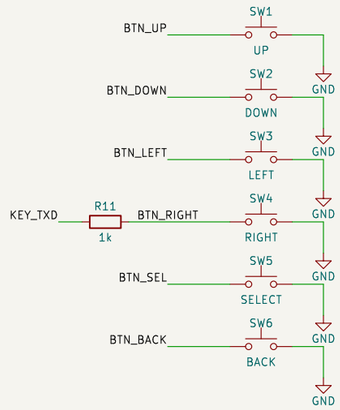


Mounting holes



Used: 3/4 GND, 5 GPIO0 SELECT (also BOOT), 7 GPIO1 pack / 2, 9 GPIO2 ARM, 10 GPIO43 RIGHT (via R11), 11 GPIO3 DOWN, 12 GPIO44 LEFT, 13 GPIO17 IR_DRV, 15 GPIO18 UP (also KEY), 16 GPIO14 BACK (I2C SCL, 2.2k pull-up on the display board).
Not connected: 1 3V3, 2 VBUS, 6/8 USB D-/D+, 14 SDA. The display board and ESP32 run from their own 18650.

D. Keys — each shorts its GPIO to GND; ESP32 internal pull-ups



R11: GPIO43 is UART TX during the boot ROM; 1k limits the current if RIGHT is held at power-up.

AA pack -> power gate -> LEDs; LED switch; pack sense; six keys; connectors

Luke Jamieson

Sheet: /
File: control_lite.kicad_sch

Title: Long-range remote LITE — control board (4 x AA NiMH)

Size: A3	Date: 2026-10-11	**Rev: G**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1